Filter Community by Minimum Sample Count

Description

Filters out taxa that are not present in a sufficient number of spatio-temporal samples (distinct dataset-age combinations). Only taxa occurring in at least 'minimum_sample_count' distinct '(dataset_name, age)' combinations are retained. This removes taxa that are present in too few interpolated time steps to provide reliable co-occurrence signal.

Usage

```
filter_community_by_minimum_sample_count(

  data_community = NULL,

  minimum_sample_count = 1

)
```

Arguments

`data_community`	A data frame containing community data in long format. Must include columns 'taxon', 'dataset_name', and 'age'.
`minimum_sample_count`	A single positive integer specifying the minimum number of distinct spatio-temporal samples (dataset-age combinations) a taxon must appear in to be retained. Default is 1 (no filtering).

Details

The function counts distinct '(dataset_name, age)' combinations per 'taxon'. Taxa with fewer combinations than 'minimum_sample_count' are removed. An error is raised if no taxa remain after filtering, which may indicate that 'minimum_sample_count' is set too high.

Value

A filtered data frame containing only taxa that appear in at least 'minimum_sample_count' distinct spatio-temporal samples. All original columns are preserved.

See Also

[filter_community_by_minimum_core_count()], [filter_community_by_minimum_proportion()], [select_top_taxa_by_group_occurrence()]